

PR-1 Data Profiler

1. Data profiling

→ Data profiling is the process of examining a dataset to understand its structure, quality, Content and statistical characteristics. It helps identify missing, ~~ff~~ values, duplicates, incorrect data type, unusual values and relationships.

2. Data science project

→ A data science project is a systematic process of problem definition, data collection, data understanding, cleaning, exploratory analysis, feature engineering, modeling when required, evaluation and communication.

3. Data frame

→ A data frame is a two-dimensional tabular data structure in pandas containing rows and columns. Different columns can contain different data types.

4. Tensor

→ A tensor is a generalization of ~~B~~ scalar, vectors and matrices to multiple dimensions. It is widely used in numerical computing and machine learning.

5. Missing values

→ Missing values are observations for which a value is not available. Common handling methods including removing appropriate record, median/mean imputation for numerical data, and mode or domain-specific imputation for categorical data.

6. Duplicate Data

- Duplicate records are repeated observations. They can bias analysis and should be identified and handled when they do not represent legitimate repeated events.

7. Data cleaning

- Data cleaning detects and corrects inaccurate, incomplete, inconsistent or duplicate data. Typical operations include missing-value handling, duplicate removal, type correction and irrelevant-column handling.

8. Exploratory Data Analysis

- EDA explores data using statistics and visualization to discover distributions, patterns, relationships and unusual observations. Univariate one variable, bivariate analyzes two and multivariate analyzes three or more.

9. Correlation

- Correlation describes the strength and direction of a linear relationship between numerical values/variables. It does not by itself establish causation.

10. Machine-learning problem

- Customer purchase can be framed as a segregation task, features include Age, Gender, Income and Orders; the target is purchaser.

11. Automated profiling Report

- Automated profiling summarizes dataset statistics, variable information, missing values, distribution, Correlation, duplicate and potential warnings

12. Data Acquisition.

- Data acquisition means collecting data from sources such as CSV files, JSON files, SQL databases and REST APIs.